

Agile Transformation Model for Large Software Development Organizations

Maarit Laanti

Nitor Delta

Aleksanterinkatu 44 A 8, 00100 Helsinki

Finland

Maarit.laanti@nitor.fi

ABSTRACT

Many organizations are going through large-scale agile transformations but they face challenges with measuring when the transformation has reached its goals or communicating what kinds of results they are expecting. Based on our experiences we developed a large-scale agile transformation model and tested it in a pilot organization. The model was well-received and taken into use as a step-by-step method to help the organization to take the next steps in their agile transformation. This paper describes our model and the results from the pilot.

CCS CONCEPTS

Software and its engineering → **Software creation and management** → **Software development process management** → **Software development methods** → **Agile software development**

KEYWORDS

Agile software development; large-scale software development; agile transformations

1 INTRODUCTION

Since the publication of Scaled Agile Framework (SAFe) and Lean Enterprise Systems and Solutions (LeSS), many agile transformations have been started around the globe [1]. We have been helping many companies to go through these kind of large-scale agile transformations over the past 10-15 years at Nokia, before SAFe and LeSS went public.

For most companies, the external pressures to be competitive together with the impact of digitalization are main reasons to adopt large-scale agility. Frameworks such as SAFe, LeSS, and others are useful starting points, because they give traditional waterfall organizations a simple cookbook-style recipe for transformation. Yet many organizations are puzzled in the midst of the transformation: are they doing the transformation right? How far are they in the transformation? What steps they should take next?

If SAFe or another framework is a recipe for the transformation, then these questions try to measure the capability of the organization, i.e. how good a cake can be baked? As we know from cooking, there are no upper limits for such capabilities and skills.

2 AGILE TRANSFORMATION MODEL

The reason why many organizations start their agile transformation is the pursuit of business success. Thus this should be the only measure for measuring agile transformations. We combined our internal knowledge of agile transformation journeys and interviewed a few trusted colleagues and asked them how they would describe the journey to large-scale agile, and how they would describe the current state they are in. Based on this we developed the categorization presented in Figure 1.

An agile transformation is based on a series of insights or Means that describe the agile mindset or reason to adopt agile behavior – like building a system in smaller increments will increase the

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

XP '17 Workshops, May 22-26, 2017, Cologne, Germany

© 2017 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-5264-2/17/05...\$15.00

<https://doi.org/10.1145/3120459.3120479>

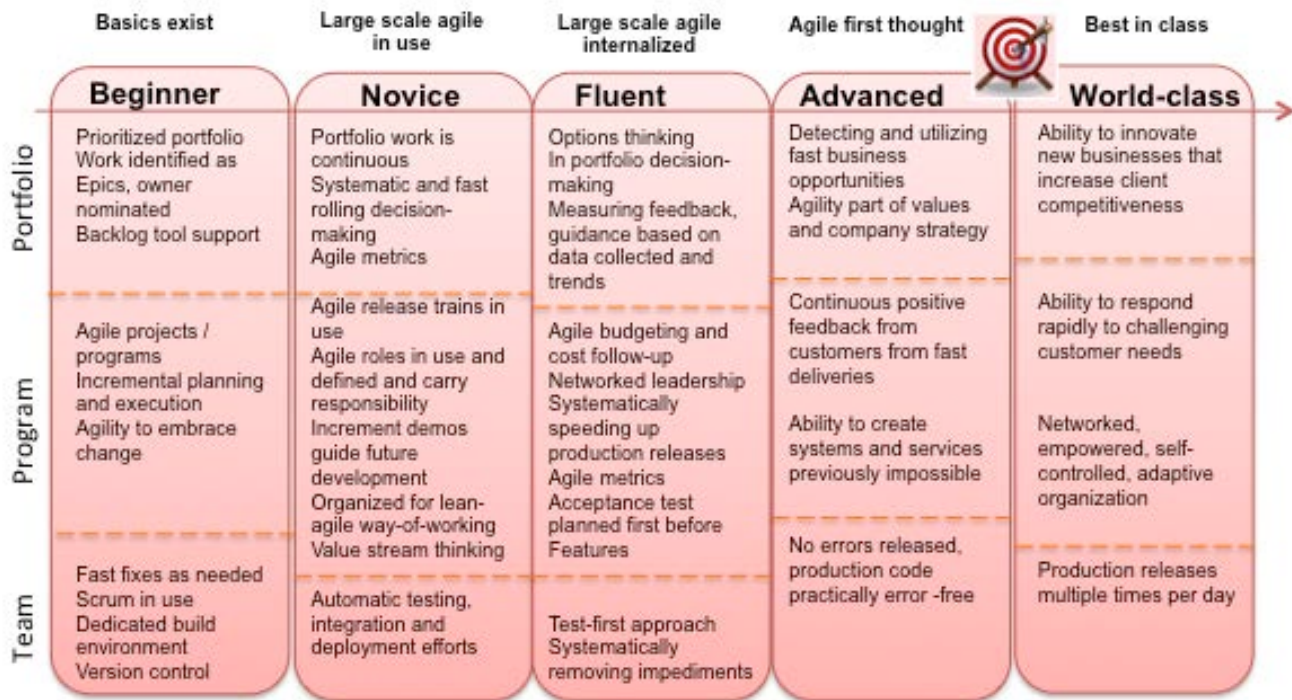


Fig. 1. Agile Transformation Model

cycle time while accumulating faster Return-On-Investment and thus the smaller increments are more productive – and Enablers that represent technical capabilities and require investments – such as investment on DevOps or automated Deployment of a Release [2]. An agile transformation is proceeding so that once the organization has reached one level they are able to gain a different insight into next level and make new investments to further advance on this path.

Most organizations would like to be among the top of their class, or to be able to compete globally. Only a few organizations however have such capabilities. For example, an organization may set a specific Goal such as “No errors released in production code” but this is not possible if the organization has significant technical debt or if the organization’s building and deployment pipeline is not very fast and automated – all of which require heavy investments on Enablers. Any misalignment between business and development (IT) may result in not achieving the business Goals and not knowing why the organization failed. There is a complex relationship between Goals, Means, and Enablers [2], and is presented in Figure 1 is actually a list of Goals.

2.1 Five Levels of the Agile Transformation Model explained

An organization on the Beginner level has understood that adaptive and flow-based models based on small batch sizes are more effective than traditional models and has decided to re-structure for large-scale agility. This is a phase where a decision is made that a certain framework is taken into. In this level the work is arranged in to backlogs and smaller batches instead of traditional projects and new tools and new roles that support agile way of working are taken into use.

An organization on the Novice level has been able to establish a flow of value. At this point the organization may abandon their traditional way of working models and take agile model as primary or only way-of-working model into use. On this level the organization is investing both making work more fluent, i.e. removing technical debt, and automation of development and testing pipeline and measuring the flow i.e. to agile metrics and radiators.

On the Fluent level the organization fully understands the benefits of agile and most of the organization’s decisions are based on quick experimentation and utilization of collected data from the markets and rapid experimental releases. These kinds of organizations are maximizing learning based on data

collected. The organization can also create a customer engagement cycle similar to Amazon [3] as the number of returning customers is its most important metric. Here the investments are made on further advancing the agility into new areas such as budgeting and business agility.

On the Advanced level the utilization of data and the process models is pushed into perfection, and Agility has become a part of company values and strategy. Agility becomes forethought rather than an afterthought. Here all the new development efforts are approached with an agile mindset, and investments are focused on new initiatives that are made agile from the start. The organization is characterized by being excited on over-exceeding its goals over and over again by better leverage provided by agility.

On the World-class level the organization has invented its own approach to agility. Investments are made on new external start-ups as the organization is already on top of its rivals and cannot or have decided not to grow in its own area.

3 EXPERIMENTING AND VALIDATION

The agile transformation model was piloted in a large client organization in the banking and insurance sector, employing 12 000 people that was going through SAFe transformation. The author worked there as a consultant, and the agile transformation model was used for gaining and understanding of the current status of the transformation and recommendations what the next steps should be. The agile transformation model was warmly welcomed as it explains to business people in plain language why they should invest in agility.

We conducted 42 interviews with 117 people in order to analyse the model, the current state, and the recommendations for the next steps. The interviews and analysis took 350 working hours. We asked in these interviews three questions:

1. In which level do you think your organization is currently?
2. In which of the five levels do you see your organization after one year's time (based on the already ongoing improvement actions)
3. In which level do you see yourself as an organization in three years time (representing ideal state)

These self-evaluations and evaluations of adjacent levels (portfolio levels evaluating programs and vice versa; programs evaluating teams and vice versa) gave the final adjusted result of the current status of agility in the whole organization. As there were multiple persons participating in most of the interviews all the answers were carefully analyzed already at that time.

After that we asked the participants to estimate where they would be after one year. We asked them to list the concrete actions that were in place already to advance the current state, and what would be further needed. This helped us to gain understanding to all development actions ongoing already, and to the ones that were planned. Finally we asked where they think they should be in three years time and what would be needed in order to get there. This helped us to gain

understanding into the systemic impediments in the organization that should be removed because these were hindering the agility.

The result was a very detailed analysis of the current state and recommended next steps reported per each portfolio and agile release train.

3.1 Coaching and teaching as main transformation tools

In some of these interviews respondents were quite happy of the current state of agility that they had accomplished. In these discussions the model had helped them to see what would be the next steps to advance their agility. In these cases we were teaching them about agile practices and how to get to the next state. This included e.g. the understanding how real-time dashboards and visibility could change how the organization is led.

In some cases the respondents already knew what they were doing and how. An example was e.g. an agile release train at an early phase, when the next steps were known. Here our role was just recording the results. In some cases the people did know what actions the others were taking but they were unwilling to follow. Examples is e.g. a case where people knew they should set-up an agile portfolio with priorities but were afraid to make the change. Here we tried to motivate them and share insights about the benefits of the new way of working. In these cases we ended up coaching the organization.

In some cases the respondents did not know what actions they should take nor cared about agile. Here we tried both to motivate and teach them to take the next steps. An example is a discussion where we discussed the importance of making releases more often. Figure 2 presents a four-square of these different kinds of interviews and presents that teaching and coaching are the main tools for making agile transformation on the individual level, when influencing individual people or groups. The idea of this quadrant model is loosely based on the Johari window [4].

3.2 Validation

During the interviews several small stories of the transformation were heard that supported the model. People in general sounded to find it very useful. "This was good, it made me think"; "This is an excellent model"; "I was positively surprised that this was not a five page long survey but simple and visual"; "I could not earlier map where we are and where we should go" and "we should do this every two months and see the progress we are making".

The critics were mainly pointing out the inductiveness of the model: as it is based on self-evaluations it will not provide any objective truth. Thus the coaching aspect as part of the interview and analysis process was considered very important. Also the

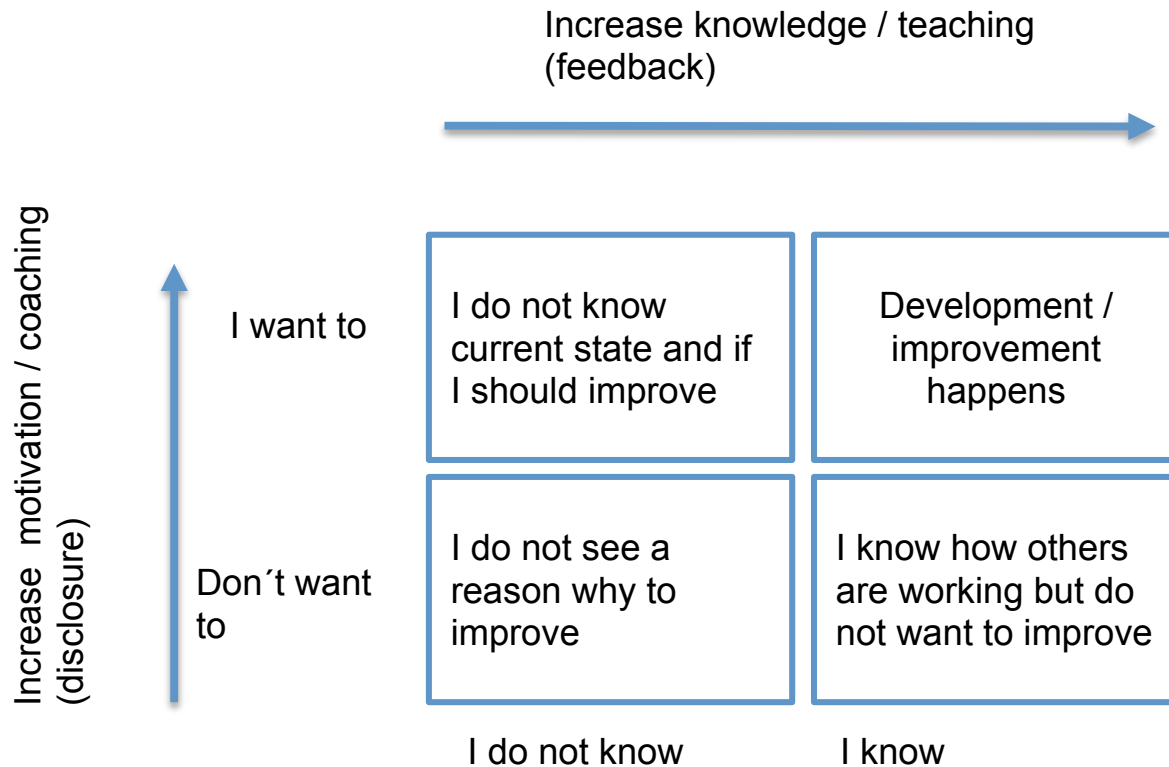


Fig. 2. Coaching and teaching as main tools for influencing people

results were not seen as comparable with each other because the model was based on self-evaluations rather than on an objective evaluation. In the end we were asked to provide our expert opinion besides the self-evaluations. This was not easy, as an external party can rely only on the information heard on interviews. Surveying organization would give wider amount of opinions and would be a way to get better results.

The final results were presented in an auditorium and widely discussed within the organization. As a result the organization decided in next steps of action to take the agility further.

4 COMPARISON TO OTHER AGILE MATURITY MODELS

Numerous agile maturity models exist but none like ours. Most of the models measure how agility is taken into use [5]. As stated by Schweigert & al [5], the focus in our model is the business value and it is descriptive, as most respondents in the survey conducted have answered to the questions “first global opinion” and “how should it be defined”. In our model the level 1 presents a situation where agile is already in use, and the next levels reflect the level of understanding, practices taken into use, and capabilities developed.

The most famous model may be Gartner’s agile maturity model [6]. This model is measuring (like most of the proposed

models) the level of agile adoption where level 0 is non-existent, level 1 is ad-hoc (some agile practices in use), level 2 means repeatable with no consistency across the organizations, level 3 is defined as the organization having documented its agile method, level 4 is quantitatively managed and level 5 is optimizing (value stream optimization) [6].

Scaled Agile Framework proposes a 1-2-3 transformation model [7] that is also targeted at getting large-scale agile methods into use. Likewise there is a commercial tool called Agility health radar that can be used to measure if an organization is following SAFe or not [8]. Yet none of these models measure what kind of results the organization is getting with its agility. We have been explaining this difference by stating that if the model is the recipe, we measure how good a cake the organization will be able to bake.

5 CONCLUSIONS

This paper has presented a tool for self-reflecting how large-scale agile transformation is progressing within an organization. The tool has been validated by experimentation and interviews. The tool has been taken into use in a few organizations that are actively using it in order to systematically advance their agile transformation. We have a plan to create a survey tool based on

the represented model in the near future in order to help our clients to make progress in their agile transformations.

An organization may have a perfect strategy, but if the execution power is a mismatch, it will not get the results it desires. Agile transformation model is a model that explains how to develop the execution power, and facilitate that discussion. It can also provide important information for developing the organization further as a system. We hope it will be useful tool for agile organization coaches.

REFERENCES

1. Dikert, Kim, Maria Paasivaara, and Casper Lassenius. "Challenges and success factors for large-scale agile transformations: A systematic literature review." *Journal of Systems and Software* 119 (2016): 87-108.
2. Kettunen, Petri, and Maarit Laanti. "Combining agile software projects and large-scale organizational agility." *Software Process: Improvement and Practice* 13.2 (2008): 183-193.
3. Norton, David. "The Current State of Agile Method Adoption." Gartner (G00163591) (2008): 1-4.
4. Luft, Joseph, and Harry Ingham. "The johari window." *Human Relations Training News* 5.1 (1961): 6-7.
5. Schweigert, Tomas, et al. "Agile maturity model: oxymoron or the next level of understanding." *International Conference on Software Process Improvement and Capability Determination*. Springer Berlin Heidelberg, 2012.
6. Norton, David. "The Current State of Agile Method Adoption." Gartner (G00163591) (2008): 1-4.
7. SAFe 4.0 Introduction, Overview of the Scaled Agile Framework® for Lean Software and Systems Engineering, Whitepaper, July 2016
8. <http://agilityhealthradar.com>, accessed 21-March-2017